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Research article

An intelligent system based on optimized ANFIS and association rules for power transformer fault diagnosis

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ABSTRACT

This research work put forward an intelligent method for diagnosis and classification of power transformers faults based on the instructive Dissolved Gas Analysis Method (DGAM) attributes and machine learning algorithms. In the proposed method, 14 attributes obtained through DGAM are utilized as the initial and unprocessed inputs of Adaptive Neuro-Fuzzy Inference System (ANFIS). In this method, attribute selection and improved learning algorithm are utilized to enhance fault detection and recognition precision. In the propounded fault detection and classification method, the most instructive attributes obtained by DGAM are selected by association rules learning technique (ARLT). Using efficient enlightening attributes and eliminating tautological attributes lead to higher accuracy and superior operation. Furthermore, appropriate training of ANFIS has significant effect on its precision and robustness. Therefore, Black Widow Optimization Algorithm (BWOA) is applied to train the ANFIS. Having excellent exploration and extraction capability, fast convergence speed and simplicity is the main reason for choosing the BWOA as the learning algorithm. Two industrial datasets are utilized to test and evaluate the performance of the put forward method. The results show that the propounded diagnosis system has high accuracy, robust performance and short run time. Selecting the most educative attributes of DGAM, training ANFIS optimally, improving the robustness of ANFIS and increasing the classification accuracy are the main contribution of this paper in the field of power transformer fault detection and classification.

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1. Introduction

Electrical three phase power transformers (TPPT) are one of the most costly and important component of modern power networks which have effective role in power system security and quality. When the transformers operate improperly and faults happen, some specific problems such as power system supply interruption which result in loss of profit from plant owners' perspective and difficulty to electrical energy buyers from consumers' perspective occurs. On the other hand, substitution of power transformer is too costly and time consuming. Hence, TPPT maintenance and monitoring their condition are a significant issue that need to be considered carefully [1–3]. Over the past few years, extensive researchers have been done in order to supervise the TPPT condition and identify the possible faults.

The various faults that appear in TPPT are generally categorized into two main groups named as interior short circuit defects and interior incipient defects. The significant fault classification methods such as inrush current measurement and Dissolved Gas Analysis Method (DGAM) [4,5] are extensively implemented for detection and classification of incipient faults. The DGAM is majorly implemented as a maintenance, surveillance, diagnostic, and control apparatus for oil-immersed devices such as power transformers [6,7]. The present DGAMs consist of Doenernberg's ratio method (DRM), gas chromatography method (GCM), IEC 60599 method, Duval triangle method (DTM), and Roger's ratio method (RRM). These approaches apply ratios of different produced gases of transformer oil for fault detection purpose. Number of these approaches compare different gas denseness to the specific amounts to assess and test the state of an oil-filled TPPT. These traditional approaches can be applied easily, but as the detection accuracy perspective, they are still have some limitations and drawbacks, and are very fragile to noises in DGAM information. Furthermore, the evaluations significantly relied on experience of experts and engineers [8]. Consequently, multiple research works have been conducted in order to enhance and improve the present DGAM's operation and accuracy.

Most of researchers believe that using computer-based DGA analysis improve the rate of accurate classification. Over the past

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Nomenclature

ANFIS	Adaptive Neuro-Fuzzy Inference System
AISA	Artificial Immune System Algorithm
AIM	Artificial Intelligence Methods
ARLT	Association Rules Learning Technique
Atr	Attributes
BWOA	Black Widow Optimization Algorithm
CFR	Composition of Feature Relevancy
C_2H_2	Acetylene
C_2H_4	Ethylene
CH_4	Methane
C_2H_6	Ethane
BP	Back Propagation
DGAM	Dissolved Gas Analysis Method
DT	Decision Tree
DTM	Duval Triangle Method
DRM	Doenernberg's Ratio Method
DEUFS	Discriminative Embedded Unsupervised Feature Selection
EDF	Energy Discharges Fault
FFNN	Feed-Forward Neural Network
IBGAFG	Improved Binary Genetic Algorithm With Feature Granulation
GA	Genetic Algorithm
GWN	Genetic-Based Wavelet Networks
GCM	Gas Chromatography Method
H_2	Hydrogen
MLPNN	Multi Layered Perceptron Neural Network
MEPSO	Modified Evolutionary Particle Swarm Optimization
MP	Membership Function
MSE	Mean Square Error
NF	No Fault
NN	Neural Network
PDF	Partial Discharge Fault
PSOA	Particle Swarm Optimization Algorithm
PNN	Probabilistic Neural Networks
OHF	Overheating Faults
OAA	One Against All
OAO	One Against One
RPLR	Resilient Propagation Learning Algorithm
RDF	Random Decision Forest
RRM	Roger's Ratio Method
RBFANN	Radial Basis Function Artificial Neural Network
SVM	Support Vector Machine
SD	Standard Deviation
TPPT	Three Phase Power Transformers
TVAC	Time Varying Acceleration Coefficient
WN	Wavelet Networks

few decades, different approaches have been introduced to improve the conventional interpretation of DGA data using Artificial Intelligence Methods (AIM) and evolutionary algorithms such as neural networks (NNs), decision tree (DT), random decision forest (RDF), fuzzy logic systems, support vector machine (SVM), and particle swarm optimization (PSO) to assess incipient fault in power transformers.

The different types of artificial NNs have been extensively implemented as classification device for TPPT fault identification [9–14]. Researchers in [9] have proposed a hybrid method, combination of artificial NN and expert system (ANNEPS), for TPPT fault detection and classification. The accuracy and performance of the introduced monitoring scheme is compared with RRM. The experimental studies have showed that ANNEPS has ability to solve the no-decision cases in RRM. Authors in [10] have proposed multi-layer Perceptron neural network (MLPNN) and artificial immune system algorithm (AISA) combination for identification and classification of faults in TPPT. The obtained results have showed that the introduced method (MLPNN-AIS) has much better performance rather than conventional DGA methods

In [11], Researchers have implemented an artificial NN-based scheme for categorizing defects of TPPTs that mixes modified PSO algorithm and artificial NN named (MPSO-NN). In this work, they established PSO algorithm to obtain a superior exploring and local search capability. Authors in [12] have put forward a fault identification scheme based on MLPNN. In [13] researchers have combined a knowledge-based system with feed-forward neural network (FFNN) to recognize and classify the faults of TPPT. Ref [14] have presented a comparative survey of the efficiency of different type of artificial NNs in recognizing internal incipient transformer defects. Artificial NNs have multiple shortcoming such as using Back-Propagation (BP) algorithm as learning algorithm that limits their implementation and as a result lower their precision and operation [15]. Hence, implementation of artificial NNs cannot be a reliable and logical approach for fault detection of oil-immersed TPPTs.

Fault detection and classification for TPPTs based on SVM are considered to be more competent and qualified as comparing to artificial NNs [16–20]. In [16] authors have applied three SVMs to identify and classify internal faults in TPPTs. In this method, the amount of five dissolved gases are used as the input of classifiers. Ref. [17] has used Genetic Algorithm (GA) in order to optimize the structure and parameters of SVM. In the introduced scheme, GA have found the optimal value of hyperplane's parameters to improve the recognition precision. In [17] and [18], new methods for prediction of ratios of dissolved gases have been propounded. In these research works, GA and PSO are utilized as optimization algorithm for determining optimal architecture of SVM. GA and PSO use simple procedure for finding global optimal answer and have been extensively used in different optimization task such training machine learning algorithms. These two methods presented in [17,18] have potential to improve the dependability and reliability of TPPT by offering practical information about the rate of failures in a small and average periods.

In [19], three classes of SVMs including one-against-all (OAA), one-against-one (OAO) and binary decision tree are utilized for fault recognition in TPPT and their performances are compared in term of recognition accuracy and run time. A multi-agent system is propounded in Ref [20] for monitoring and assessing the TPPT condition. In this research work, the performance of SVM in fault detection in TPPT is compared with radial basis function neural network (RBFNN) in term of accuracy, robustness and generalization. The obtained result through computer software simulation have revealed that SVM has better operation than RBFNN. However, SVM has some natural and inherent difficulties such as computational burden that restrict its performance [21].

In [22–24], fuzzy logic based schemes have been utilized for TPPT fault recognition. For example, in [22] authors have developed a knowledge-based system for fault diagnosis of TPPT. In this method, researchers have utilized type-2 fuzzy system as intelligent classifier and have used oil chemical characteristics and

Table 1
Attributes utilized for fault diagnosis of TPPT.

Atr_1	H_2	Atr_8	C_2H_4/C_2H_6
Atr_2	CH_4	Atr_9	C_2H_6/C_2H_2
Atr_3	C_2H_4	Atr_{10}	C_2H_2/CH_4
Atr_4	C_2H_6	Atr_{11}	$CH_4/(CH_4 + C_2H_2 + C_2H_4)$
Atr_5	C_2H_2	Atr_{12}	$C_2H_4/(CH_4 + C_2H_2 + C_2H_4)$
Atr_6	C_2H_2/C_2H_4	Atr_{13}	$C_2H_2/(CH_4 + C_2H_2 + C_2H_4)$
Atr_7	CH_4/H_2	Atr_{14}	$H_2 + CH_4 + C_2H_2 + C_2H_4 + C_2H_6$

value of dissolved gases as the input vector. Authors in Ref [23] has proposed a rule-based scheme that employ fuzzy inference motor for fault recognition in TPPTs. Another research work using fuzzy rules and fuzzy inference motor for fault identification in TPPT is presented in Ref [24], which establish multiple rules corresponding to each internal fault. Several similar methods using Wavelet networks (WN), genetic-based wavelet networks (GWNs), and grey clustering analysis are presented in by different researchers [25–27]

Based on the previously performed researches in the field of fault identification in TPPTs, there are two important issues that need to be addressed. These important issues are selection of the most educational attributes (Atr) from DGAM information, and type of classifier. In this paper, we implemented association rules learning technique (ARLT) to select the most instructive attributes and minimize the input data dimension. In addition, we used adaptive neuro-fuzzy inference system (ANFIS) as the main classifier. The excellent operation of ANFIS in classifications, fault detection and pattern recognition tasks have been proved by many research works [28,29]. Similar to ANNs, ANFIS uses Gradient-based methods as learning algorithm to determine the weights and other parameters, and has its inherent problems. To improve the performance of ANFIS, we used novel learning algorithm based on Black Widow Optimization Algorithm (BWOA) that has been introduced in [30]. More details about proposed method is offered in the third section.

This article is organized as follows. The investigated dataset is introduced in second section. Section three describes the proposed system. Section four shows some simulation results and finally section five concludes the paper.

2. Dataset

Interpretation of information obtained via DGAM is one of the common tools for the identification and classification of internal faults in TPPT. The different gases that are produced and decomposed in oil of the TPPTs due to different internal faults and conditions are ethane (C_2H_6), acetylene or ethyne (C_2H_2), ethylene or ethene (C_2H_4), methane or march gas (CH_4), and hydrogen (H_2) [31,32]. Different interpretation methods use the proportion of these generated gases for detection and classification of internal defects. Different internal defects produce various levels of heat that result in different chemical response inside the TPPT's insulating fabrics and therefore generate equivalent dissolved gases. Various proportions of these gases could be applied to better faults detection as well as their pure amount. The 14 input variables used in the proposed method are listed in Table 1. The conditions of transformer are classified into four classes named partial discharge fault (PDF), energy discharges fault (EDF), overheating faults (OHF) and no fault (NF).

In the experiments and for accurate and practical investigation purpose, we have used two practical dataset to assess the performance of the propounded method. The dataset 1 is collected by Iran-Transfo Company chemical laboratory in 2017. Iran-Transfo Company is located in Zanjan, Iran and is the largest transformer production company which produce different types of power transformers under license of Siemens in Iran [33]. More details regarding Iran-Transfo Company and Dataset 1 can be found in [33]. The Dataset 2 is collected in Damavand power substations located in Tehran, Iran in 2017 [34]. The number of instances in each dataset for the three classes and no-fault case are presented in Table 2. In addition, Tables 3 and 4 show some of the actual dissolved gas analysis for dataset 1 and dataset 2, respectively.

Table 2
Types of investigated faults.

Fault type	No. samples	
	Dataset 1 [33]	Dataset 2 [34]
ODF	105	70
EDF	105	85
OHF	133	79
NF	197	108
	Total samples = 540	**Total samples = 342**

Table 3
Some samples of dataset 1 [33].

H_2	CH_4	C_2H_4	C_2H_6	C_2H_2	Fault type
98	34	3	0	1	NF
120	963	680	74.5	311	NF
22.3	0.001	0.001	14	0.001	NF
14.7	3.9	10.5	2.7	0.2	NF
8.5	7.2	4.3	3.9	3.51	NF
240	20	5	28	96	PD
650	53	34	20	0	PD
117	18	0.001	1	0.001	ED
858	1324	208	2793	7672	ED
79	21	12	10	26	ED
82	622	327	2485	0.001	OHF
231	3997	1726	5584	0.001	OHF
65	61	16	143	3	OHF
28	87	65	40	0.26	OHF
1078	1594	2007	229	25	OHF
294	959	1796	292	57	OHF

Table 4
Some samples of dataset 2 [34].

H_2	CH_4	C_2H_4	C_2H_6	C_2H_2	Fault type
11	12.5	73	31	22	NF
3	1	1	2	1	NF
26	30.5	23	22	0.1	NF
0.001	19.3	0.001	57.2	0.001	NF
0.001	4	0.001	11	0.001	NF
557	397	359	241	269	NF
2496	1465	5001	4599	7021	NF
1023	95	4	71	231	PD
9032	785	12	219	36	PD
1565	93	34	47	0.001	ED
297	240	14	159	138	ED
2400	774	315	4462	647	ED
42	97	157	603	0	OHF
73	520	140	1995	6	OHF
199	770	217	1508	72	OHF
2754	16 615	3657	31 476	613	OHF
29.6	115	135	43.5	0.15	OHF
131	142	118	3	0.001	OHF

3. Proposed method

In this section, the developed method for power transformer faults detection and classification is presented with details. In scheming of computer-based classification systems, the most significant issue is appropriate application of input features and classifiers [35–40]. Consideration has been given to this fact in the developed method.

Fig. 1 shows the conventional ANFIS-based faults classification system in TPPTs. In this system, raw DGAM data is fed to ANFIS.

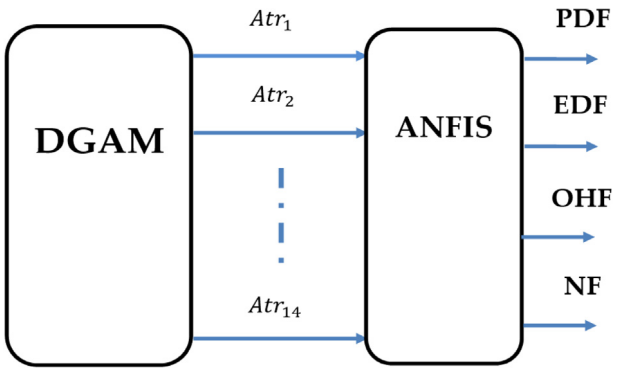


Fig. 1. Conventional ANFIS-based faults detection and classification system in TPPTs.

The attributes (Atr) of raw DGAM data presented in Table 1. There are 14 attributes in raw DGA data.

Weak educational attributes play as artificial noise that result in poor performance and low classification accuracy. Hence, the uninformative attributes must be removed from data bank and only effective attributes must be utilized as the input. In our study problem, by selecting the best collection of attributes, the volume of input data to ANFIS will decrease from 14 (in unprocessed data state) to smaller number (using chosen attributes). In Fig. 2, M attributes out of 14 original attributes have been picked, and Atr_s indicates the selected attribute where $i = 1, 2, \ldots, M$ and $M < 14$. In the proposed method, we used ARLT as the attribute selection approach. The ARLT is a rule-based machine learning method for discovering interesting relations between variables in large databases. The ARLT analysis is a technique to uncover how items are associated to each other. There are three generalized approach to assess association involving support, confidence and lift. The ARLT identifies frequent if-then associations called association rules which consists of an antecedent (if part) and a consequent (then part). More details regarding AR and its formulation can be found in [41].

In ANFIS, the value of initially unknown parameters determine the accuracy of classification task. These initially unknown parameters named antecedent and conclusion parameters are shown by Fig. 3. Conventional ANFIS uses gradient-based algorithm as learning algorithms, which have some natural drawbacks such as low convergence speed and high chance of getting trapped in local minima. Moreover, the random initial solution in gradient-based algorithm determine the success of algorithm. The weak and inappropriate initial solution will lead to local minima and failure in training the ANFIS [42].

Nature-based optimization algorithms can tackle various nonlinear and complicated engineering problems because of their flexibility, resilience and no need to gradient information. The BWOA is one of the newest evolutionary nature-based optimization algorithms, which is modeled and developed based on the unique mating behavior of black widow spiders. Like other optimization algorithms, BWOA starts with random population in search space and uses some strategies to approach the global answer. More details regarding BWOA can be found in [30]. The general structure of proposed method and its main steps is shown by Fig. 4. In the proposed method, ANFIS is formed by using BWOA in order to classify TPPT faults.

A sample spider in the proposed method is shown in Fig. 5. In this example, the type of membership functions (MP) for input variables is $\pi - Shape$. The type of membership functions depends on problem characteristics and should be selected by designer. In this figure, $a_{1,1}$ $b_{1,1}$ $c_{1,1}$ show the parameters of first inputs first membership function, $a_{1,2}$ $b_{1,2}$ $c_{1,2}$ show the parameters of first inputs second membership functions parameters and so on. Also $a_{N,M}$ $b_{N,M}$ $c_{N,M}$ show the parameters of Nth inputs Mth membership functions parameters. The number of input variables (N) depends on problem. In this method, the number of input variables determined by feature selection algorithm. Also the number of membership functions describing each input (M) determined by designer. Furthermore p_1 q_1 r_1 show the conclusion parameters. The number of conclusion parameters depend on number of fuzzy rules. In this figure, K shows the number of fuzzy rules.

4. Simulation results

In this section, the accuracy and operation of the developed method is investigated where the practical and real world data described in section two [33,34] are used for this purpose. In the simulations using MATLAB environment, 50% of data is utilized for training and 50% is utilized for testing purpose. All the mentioned results in the following subsections are the average of 40 independent runs. Having lower standard deviation (SD), shows that classifier has robust performance and is more reliable.

4.1. Performance of the proposed method

In this study, we used ARLT to select the best attributes obtained by DGAM in order to improve the recognition accuracy and shorten the run time. The collected database have four classes. These are partial discharges, energy discharges, overheating and no fault classes. Using AR, we found large item sets of partial discharges, energy discharges, overheating and no fault classes given as follows:

Input: 1, 2, 4 and 12 (large item sets for no fault class)
Input: 5, 8 and 12 (large item sets for partial discharges class)
Input: 2, 5 and 8 (large item sets for energy discharges class)
Input: 1, 3, 4 and 11 (large item sets for overheating class)

According to this large item set, we can say that Atr_1, Atr_2, Atr_4 & Atr_{12} input parameters already can define no fault class, Atr_5, Atr_8 & Atr_{12} input parameters already can define partial discharges class, Atr_2, Atr_5 & Atr_8 can define energy discharges class and Atr_1, Atr_3, Atr_4 & Atr_{11} input parameter can define overheating class. These features are the most important and informative features for transformer fault detection and classification problem. Therefore, we only used these eight features as inputs of ANFIS. These features are:

Attribute 1 (H_2)
Attribute 2 (CH_4)
Attribute 3 (C_2H_4)
Attribute 4 (C_2H_6)
Attribute 5 (C_2H_2)
Attribute 8 (C_2H_4/C_2H_6)
Attribute 11 ($CH_4/(CH_4 + C_2H_2 + C_2H_4)$)
Attribute 12 ($C_2H_4/(CH_4 + C_2H_2 + C_2H_4)$).

These eight attributes have been used as the best input of ANFIS. Also the antecedent and conclusion parameters of ANFIS selected by BWOA (BWOA-ANFIS). The obtained results are shown in Tables 5 and 6. In these tables, Gradient-ANFIS represent the standard ANFIS with traditional learning algorithm and BWOA-ANFIS refers to ANFIS with proposed learning algorithm where its parameters are found by BWOA. In addition, to explore the effectiveness of ARLT, we utilized various feature collection as input of ANFIS. For the dataset 1, it can be seen that 97.82% classification precision is gained by BWOA-ANFIS and original five features, 96.71% classification precision is acquired by BWOA-ANFIS and feature 6 to 14, 98.51% classification precision is achieved by BWOA-ANFIS and whole 14 features and

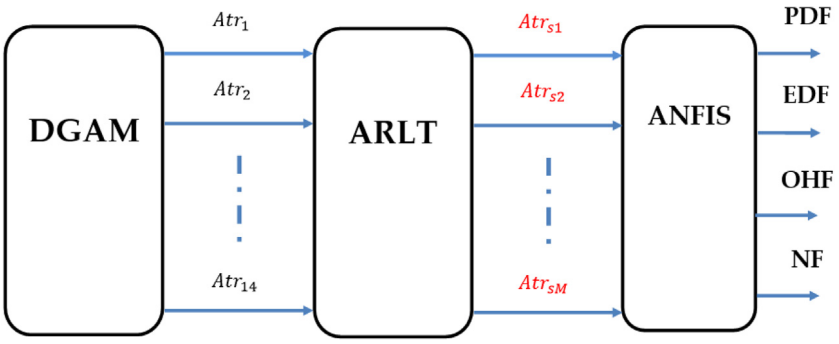


Fig. 2. Attribute selection in the propounded fault classification scheme.

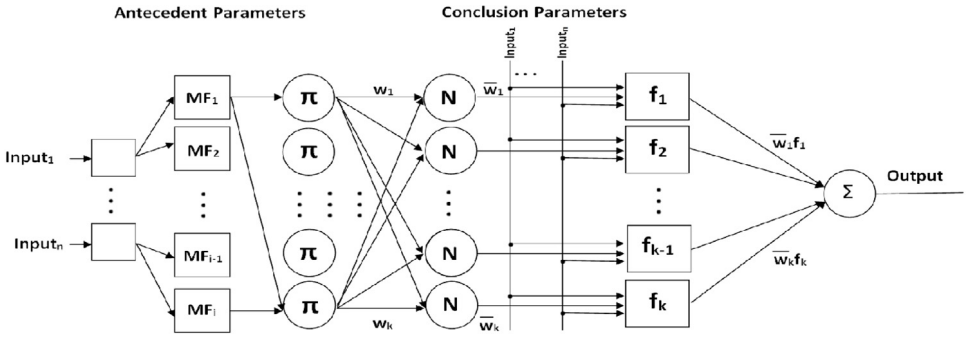


Fig. 3. ANFIS structure [42].

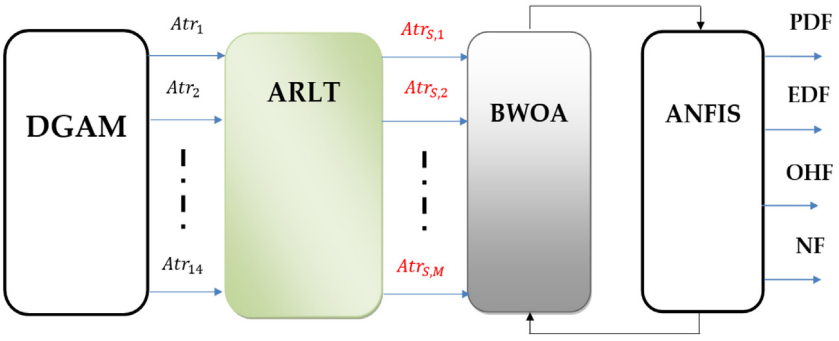


Fig. 4. The general framework of proposed method.

$$[a_{1,1}\ b_{1,1}\ c_{1,1}\ a_{1,2}\ b_{1,2}\ c_{1,2}\ \ldots\ldots\ a_{N,M}\ b_{N,M}\ c_{N,M}\ ;\ p_1\ q_1\ r_1\ \ldots\ldots\ p_K\ q_K\ r_K]$$

Antecedent parameters Conclusion parameters

Fig. 5. Sample spider in the propounded fault detection method.

99.67% classification precision is obtained by proposed method. Similar results have been achieved for dataset 2. The obtained results show that the proposed method (BWOA-ANFIS and best feature collection) has the best recognition accuracy, 99.67%. This experiment indicates and proves the significance of attribute selection and learning algorithm.

The value of standard deviation of classifiers in 40 independent runs mentioned in Tables 5 and 6. In this tables, the value of standard deviation for the developed method is equal to zero, (SD $= \pm 0.00$) which proves the excellent and robust performance of the developed method. In addition, this fact indicates that BWOA could find the best parameters of ANFIS in different implementations. In contrast, gradient-based learning algorithm has high standard deviation value commonly, which means that result in different outputs at different implementations. Hence, gradient-based learning algorithms are not secure and dependable approaches. Figs. 6–8, illustrate these facts. Fig. 6 compares the BWOA and gradient-based learning algorithm with respect to convergence speed and classification accuracy. In these plots, the x-axis shows the iteration number in training stage and the y-axis indicates the value of defined fitness function, mean square error (MSE). It has been observed that BWOA has outstanding performance in comparison with gradient-based learning algorithm.

Figs. 7 and 8 illustrate the operation of BWOA and gradient-based learning algorithms in different four independent implementation. The obtained results indicate that BWOA have robust and reliable operation at various implementations and could train the ANFIS correctly. These plots indicate that BWOA-ANFIS may

Table 5
Performance of developed scheme for dataset 1.

Input attributes	No. input	Classifier	Accuracy (%)	SD	Run time (s)
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	Gradient-ANFIS	97.00	±4.62	8.72
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	BWOA-ANFIS	97.82	±0.74	1.04
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	9	Gradient-ANFIS	95.00	±4.76	9.63
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	9	BWOA-ANFIS	96.71	±0.81	1.34
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	Gradient-ANFIS	96.66	±3.31	10.72
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	BWOA-ANFIS	98.51	±0.29	1.64
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8$ Atr_{11}, & Atr_{12}	8	Gradient-ANFIS	98.66	±1.48	9.11
$\boldsymbol{Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8}$ $\boldsymbol{Atr_{11}}$, & $\boldsymbol{Atr_{12}}$	**8**	**BWOA-ANFIS**	**99.62**	±0.00	**1.16**

Table 6
Performance of developed scheme for Dataset 2.

Input attributes	No. input	Classifier	Accuracy (%)	SD	Run time (s)
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	Gradient-ANFIS	97.40	±4.57	6.73
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	BWOA-ANFIS	97.96	±0.68	0.74
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	9	Gradient-ANFIS	95.83	±4.62	6.98
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	9	BWOA-ANFIS	96.39	±0.73	1.06
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	Gradient-ANFIS	96.88	±3.46	7.28
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	BWOA-ANFIS	98.53	±0.26	1.17
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8$ Atr_{11}, & Atr_{12}	8	Gradient-ANFIS	98.96	±1.26	5.49
$\boldsymbol{Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8}$ $\boldsymbol{Atr_{11}}$, & $\boldsymbol{Atr_{12}}$	**8**	**BWOA-ANFIS**	**99.48**	**±0.00**	**0.82**

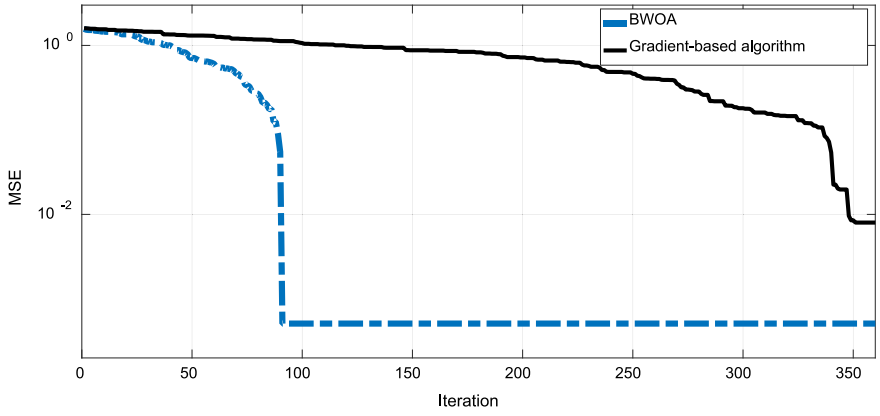


Fig. 6. BWOA and gradient-based learning algorithms comparison with regard to their accuracy and convergence speed in training stage.

result in unique performance at various implementations and is dependable classifier. At the same time, gradient-based ANFIS has met to different answers in considerably different number of epochs.

In the developed method, Gaussian membership function is implemented as for defining the input variables in fuzzy language. In the developed fuzzy logic system, four fuzzy rules are built using three membership functions for each input variable. Two control parameters including center and spread (width) determine the Gaussian membership function's behavior. We have eight inputs (selected attributes from DGAM information) and each input variable has been defined through two Gaussian membership function. Hence, there will be $8 \times 2 \times 3 = 48$ antecedent parameters in the developed fuzzy system. The quantity of conclusion parameters of every fuzzy rule is one unit more than the quantity of input variables. In the framework of developed fuzzy system, with eight selected attributes, the quantity of conclusion parameters of every fuzzy rule will be nine. Hence, there will be $4 \times 9 = 36$ antecedent parameters in the framework of developed fuzzy system. Accordingly, $48 + 36 = 84$ parameters' optimal value in total are discovered by BWOA.

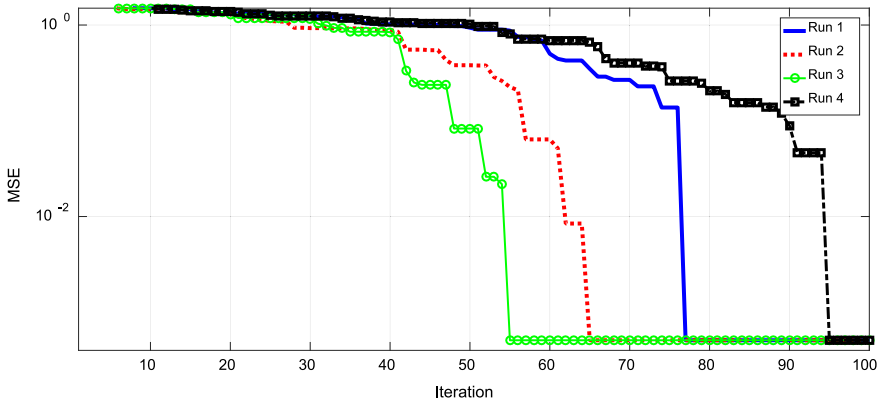


Fig. 7. Operation of BWOA at different implementations.

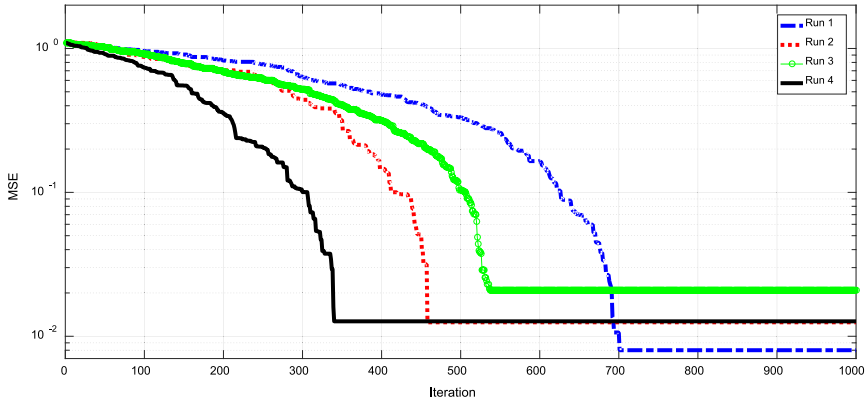


Fig. 8. Operation of gradient-based algorithm at different implementations.

As an example, MPs determined by BWOA and gradient-based learning algorithm for the 1st and 2nd input variables are outlined by Figs. 9 and 10. It can be observed that MPs established by various algorithms have led to different parameters. These diminutive differences between MPs lead to diverse recognition accuracies.

The convergence speed of BWOA-based training process are is by Figs. 11 and 12 as an example. It is observed that the developed learning algorithm has an extremely good convergence speed. As demonstrated by Figs. 11 and 12, ANFIS parameters step by step changed from first iteration to final predetermined maximum number of iterations, and exhibited no remarkable change after iteration 57. In fact, after 57 iterations, the algorithm has arrived to the best point. Same outputs are achieved for other control parameters.

4.2. Comparison with other methods

Due to high importance of TPPT in power system's security and finical issues, over the past decades, many research works have been conducted in this area and different methods have been introduced for continuous monitoring of TPPT status. Straightforward comparison with other developed schemes is a complicated task, because there is not exist a single benchmark data bank available for investigations and comparison. In the conducted researches by different research workers, different number of training and testing cases, different number of patterns and different criteria will lead to various operations. These facts bring about difficulties for straightforward numerical and statistical comparison.

For comparison, we applied two recently introduced method on our datasets and compared with our method. In [8] a new method has introduced for fault diagnosis in TPPT, which consist of two major phases, named (Two-ADOPT). In this study, authors have used binary version of the multi-objective PSO (MOPSO) algorithm. In the first stage of the developed scheme, MOPSO selects the effective subset of attributes in accordance with each fault category of TPPTs. Table 7 lists the best attributes in accordance with each fault category. After that, in the following step, ensemble-learning systems have been applied to classify actual faults of TPPTs.

Table 7
Selected attributes for each fault category.

Fault category	Chosen attributes
NF	$Atr_1, Atr_2, Atr_3, Atr_5, Atr_6, Atr_7, Atr_9$ & Atr_{14}
PDF	$Atr_1, Atr_4, Atr_5, Atr_6,$ & Atr_{12}
EDF	$Atr_2, Atr_4, Atr_5, Atr_6, Atr_{10}$ & Atr_{11}
OHF	$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_7, Atr_8$ & Atr_{11}

In [11], authors have utilized an artificial NN-based scheme for classifying faults in TPPT called (MPSO-NN). In the developed scheme, authors have proposed and applied some modification in PSO algorithm to improve its local searching behavior and enhance its accuracy. In this method, first five attributes of DGAM ($Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5) have been used as the input of artificial NN.

The obtained results using various methods are listed in Table 8. In this table, the recognition accuracy, run time based on second and SD of recognition accuracy of mentioned methods on Dataset 1 and Dataset 2 are listed. It can be seen that best recognition accuracy for both dataset is achieved by the proposed method. The SD of various methods show that the proposed

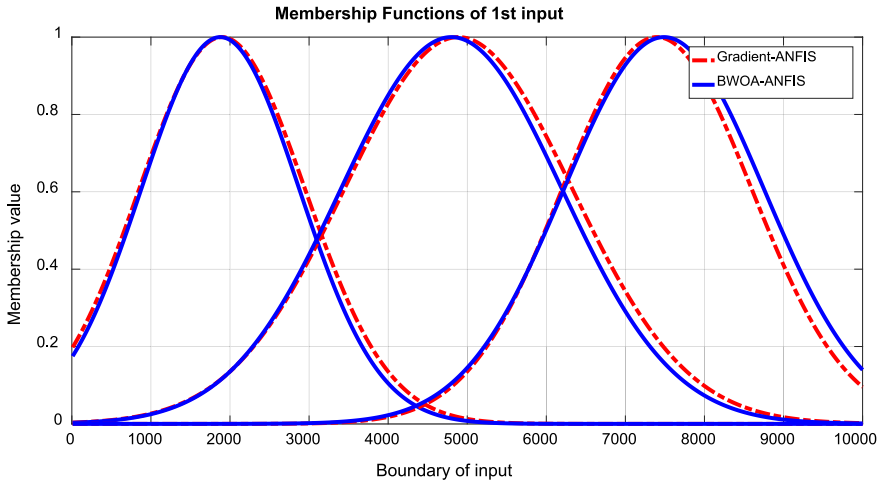


Fig. 9. Defined MPs for the 1st input using BWOA and gradient-based algorithm.

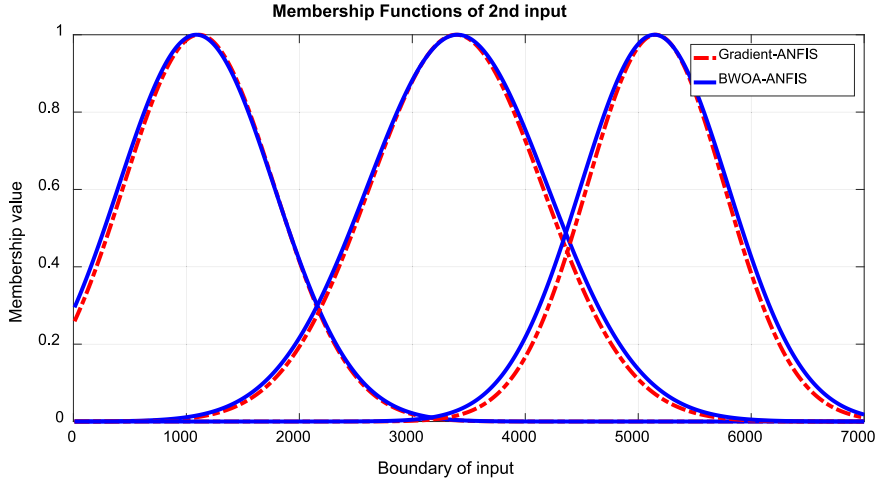


Fig. 10. Defined MPs for the 2nd input using BWOA and gradient-based algorithm.

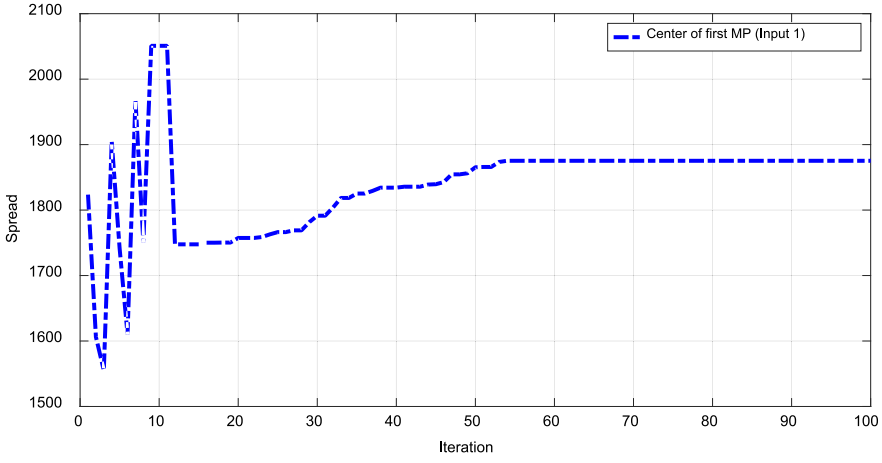


Fig. 11. Convergence of BWOA in searching for optimal center.

method has more robust performance compared to other methods. The SD of the proposed method is zero and it means that the proposed method can recognize the fault type with high accuracy in each run. Also the run time of the proposed method is significantly better rather than other methods. Higher recognition accuracy, lower run time and robust performance during different runs are advantages of the proposed method in comparison with other methods.

For further investigations, we compared the performance of the proposed method with other methods in the literature. Table 9 compares different methods in case of the accuracy and the classifier type. In each study, different databases have been

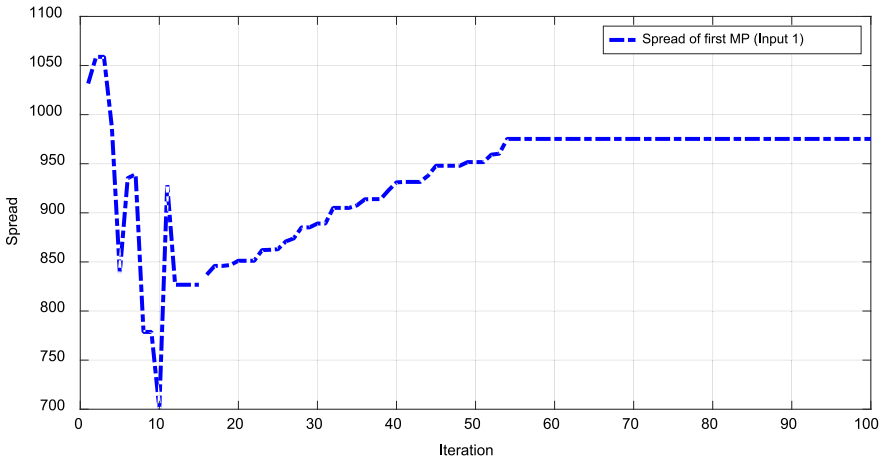


Fig. 12. Convergence of BWOA in searching for optimal spread.

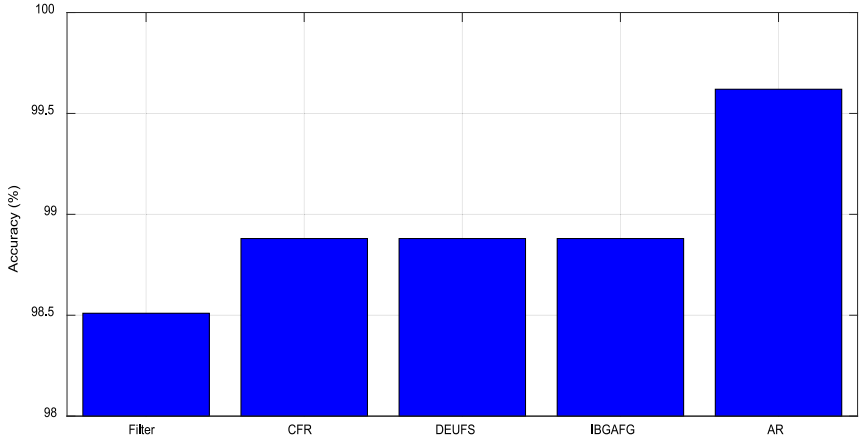


Fig. 13. Effect of attribute selection approach on performance of classifier.

Table 8
Accuracy of the developed scheme and comparison with other schemes.

Classifier	Accuracy (%)		Run time (s)		SD	
	Data 1	Data 2	Data 1	Data 2	Data 1	Data 2
2-ADOPT	98.88	98.95	27.22	21.65	±0.41	±0.37
MPSO-NN	98.15	98.43	13.75	9.60	±0.52	±0.48
RRM	94.44	96.49	–	–	–	–
DRM	95.55	97.07	–	–	–	–
BWOA-ANFIS	**99.62**	**99.48**	**1.16**	**0.82**	**±0.00**	**±0.00**

used and in this table we just compared the best obtained result which mentioned in each study. Comparison of the results shows that the proposed method is superior to other methods.

4.3. Assessment of attribute selection approach

The operation of the propounded attribute selection approach, ARLT, is compared with other attribute selection methods, for investigating the capability of the ARLT. In this respect, filter method [47], composition of feature relevancy (CFR) [48], discriminative embedded unsupervised feature selection (DEUFS) [49] and improved binary genetic algorithm with feature granulation (IBGAFG) [50] are considered. The obtained results are shown by Table 10 and Fig. 13. It is observed that the developed method in this study, ARLT+BWOA-ANFIS, has much higher recognition precision than other recognition approaches. In all developed classifiers, the ANFIS is trained by BWOA.

4.4. Comparison with various classifier

In this subsection, the accuracy of the developed machine-learning algorithm, ANFIS, has been compared with other machine-learning algorithm. In this experiment, probabilistic neural networks (PNN), RBFNN, MLPNN with Levenberg–Marquardt algorithm (LM) learning algorithm and SVM are considered. In this experiment, we used different attributes subsets. It can be seen from Table 11 that the developed method based on ANFIS has much higher classification accuracy than other machine-learning algorithms.

5. Conclusion

In this research work, a reliable and accurate scheme put forward for TPPT fault diagnosis. We used two practical datasets to evaluate and assess the operation of the developed approach. A couple of computer software experiments conducted to test the operation of the introduced scheme. In the first simulation, ARLT utilized to choose the most efficient attributes obtained by DGAM. The ARLT must select the most separating and enlightening attributes in order to minimize the input variable numbers and decrease the computational burden. In addition, BWOA implemented to find the optimal parameters of ANFIS. ANFIS, with optimal structure and selected attributes was able to precisely identify the faults in TPPTs, 99.62% precision for dataset 1 and 99.48% accuracy for dataset 2.

Table 9
A summary of different classification algorithms with their reported results.

Ref	Year	Method	Accuracy (%)
[43]	1993	Evolutionary programming-artificial NN (EPANN)	95.00
[44]	1998	Self-organizing polynomial network (SOPN)	97.68
[9]	2000	Artificial NN-expert system (ANNEPS)	90.95
[25]	2003	Genetic wavelets network (GWN)	96.19
[45]	2009	Genetic programming-k-NN (GP-KNN)	92.11
[46]	2012	Rough Set Theory	81.25
[19]	2013	Support Vector Machine (SVM)	92.00
[10]	2014	Immune Neural Network	86.30
[11]	2016	MLP neural network with optimal structure	99.00
[8]	2017	Two-ADOPT	99.00
–	–	**Proposed method (AR + BWOA-ANFIS) on dataset 1**	**99.62**
–	–	**Proposed method (AR + BWOA-ANFIS) on dataset 2**	**99.48**

Table 10
Comparison the performance of different feature selection methods.

Classifier	Selected features	Accuracy (%)	
		Dataset 1	Dataset 2
Filter + BWOA-ANFIS	$Atr_1, Atr_2, Atr_5, Atr_8, Atr_9$ & Atr_{14}	98.51	98.24
CFR + BWOA-ANFIS	$Atr_1, Atr_3, Atr_4, Atr_8, Atr_9, Atr_{10}$ & Atr_{14}	98.88	98.83
DEUFS + BWOA-ANFIS	$Atr_1, Atr_2, Atr_4, Atr_5, Atr_8, Atr_{11}$, Atr_{13}, Atr_{12} & Atr_{14}	98.88	98.83
IBGAFG + BWOA-ANFIS	$Atr_1, Atr_3, Atr_4, Atr_8, Atr_9, Atr_{11}$, Atr_{12} & Atr_{14}	98.88	98.83
Proposed method (ARLT + BWOA-ANFIS)	$\boldsymbol{Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8}$ $\boldsymbol{Atr_{11}}$, & $\boldsymbol{Atr_{12}}$	**99.62**	**99.48**

Table 11
Performance of proposed method.

Input features	No. input	Classifier	Accuracy (%)	
			Dataset 1	Dataset 2
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	PNN	94.44	92.39
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	RBF	96.29	94.15
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	MLP (LM)	95.92	93.56
$Atr_1, Atr_2, Atr_3, Atr_4$ & Atr_5	5	SVM	96.66	94.15
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}$, Atr_{12}, Atr_{13} & Atr_{14}	9	PNN	93.70	90.05
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}$, Atr_{12}, Atr_{13} & Atr_{14}	9	RBF	95.18	92.29
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}$, Atr_{12}, Atr_{13} & Atr_{14}	9	MLP (LM)	94.81	92.39
$Atr_6, Atr_7, Atr_8, Atr_9, Atr_{10}, Atr_{11}$, Atr_{12}, Atr_{13} & Atr_{14}	9	SVM	95.55	92.98
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9$ $Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	PNN	94.81	93.56
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9$ $Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	RBF	96.66	97.07
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9$ $Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	MLP (LM)	95.18	95.90
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_6, Atr_7, Atr_8, Atr_9$ $Atr_{10}, Atr_{11}, Atr_{12}, Atr_{13}$ & Atr_{14}	14	SVM	97.03	97.07
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8\ Atr_{11}$, & Atr_{12}	8	PNN	97.40	96.49
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8\ Atr_{11}$, & Atr_{12}	8	RBF	98.14	98.24
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8\ Atr_{11}$, & Atr_{12}	8	MLP (LM)	98.14	97.66
$Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8\ Atr_{11}$, & Atr_{12}	8	SVM	98.51	98.83
$\boldsymbol{Atr_1, Atr_2, Atr_3, Atr_4, Atr_5, Atr_8\ Atr_{11}}$, & $\boldsymbol{Atr_{12}}$	**8**	**Proposed method**	**99.62**	**99.48**

In the next experiments, the operation of the developed scheme compared with other methods introduced in the publications. In this experiment, we applied two recently introduced methods called Two-ADOPT and MPSO-NN on our datasets. The obtained results showed that the proposed method had higher accuracy, lower run time and more robust performance in comparison with other methods. Also we compared the performance of the proposed method with other methods introduced in literature. We compared the recognition accuracy of our method with other methods accuracy reported in the publications. Comparisons showed that the proposed method had higher recognition accuracy.

In the third experiment, the operation of the ARLT compared with other attribute selection approaches. The obtained results showed that ARLT had better performance in comparison with other methods. In the last experiment, the performance of the proposed method compared with different classifiers. It observed that the developed scheme has led to higher accuracy in comparison with other schemes.

According to the obtained results, we can conclude that the attribute selection has high influence on fault detection and classification accuracy. In addition, the type of classifier such as ANFIS, SVM, RF or artificial NN is very important issue and it is critically necessary to consider this issue during the designing a fault diagnosis system in TPPTs. The smart combination of

attribute selection algorithm (ARLT), optimized classifier (BWOA-based ANFIS), improvement of ANFIS accuracy and robust performance are the main contribution of the developed scheme. The obtained results through MATLAB simulations demonstrated that the proposed diagnosis system has robust operation, high precision and brief implementation time.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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